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### 11.8.4 Fundamental frequency and overtones

Let us now keep the rigid boundaries at $x = 0$ and $x = L$ and produce a standing waves by wiggling the string (as in plucking strings in a guitar). Standing waves with a specific wavelength are produced. Since, the amplitude must vanish at the boundaries, therefore, the displacement at the boundary must satisfy the following conditions

$$y(x = 0, t) = 0 \text{ and } y(x = L, t) = 0 \qquad (11.68)$$

Since the nodes formed are at a distance $\frac{\lambda_n}{2}$ apart, we have $n\left(\frac{\lambda_n}{2}\right) = L$, where $n$ is an integer, L is the length between the two boundaries and $\lambda_n$ is the specific wavelength that satisfy the specified boundary conditions. Hence,

$$\lambda_n = \left(\frac{2L}{n}\right) \qquad (11.69)$$

*What will happen to wavelength if n is taken as zero? Why is this not permitted?*

Therefore, not all wavelengths are allowed. The (allowed) wavelengths should fit with the specified boundary conditions, i.e., for n = 1, the first mode of vibration has specific wavelength $\lambda_1 = 2L$. Similarly for n = 2, the second mode of vibration has specific wavelength

$$\lambda_2 = \left(\frac{2L}{2}\right) = L$$

For n = 3, the third mode of vibration has specific wavelength

$$\lambda_3 = \left(\frac{2L}{3}\right)$$

and so on.

The frequency of each mode of vibration (called natural frequency) can be calculated. We have,

$$f_n = \frac{v}{\lambda_n} = n\left(\frac{v}{2L}\right) \qquad (11.70)$$

The lowest natural frequency is called *the fundamental frequency*.

$$f_1 = \frac{v}{\lambda_1} = \left(\frac{v}{2L}\right) \qquad (11.71)$$

The second natural frequency is called *the first over tone*.

$$f_2 = 2\left(\frac{v}{2L}\right) = \frac{1}{L}\sqrt{\frac{T}{\mu}}$$

The third natural frequency is called *the second over tone*.

$$f_3 = 3\left(\frac{v}{2L}\right) = 3\left(\frac{1}{2L}\sqrt{\frac{T}{\mu}}\right)$$

and so on.

Therefore, the $n^{th}$ natural frequency can be computed as integral (or integer ) multiple of fundamental frequency, i.e.,

$$f_n = nf_1, \quad \text{where } n \text{ is an integer} \qquad (11.72)$$

*If natural frequencies are written as integral multiple of fundamental frequencies, then the frequencies are called harmonics*. Thus, the first harmonic is $f_1 = f_1$ (the fundamental frequency is called first harmonic), the second harmonic is $f_2 = 2f_1$ , the third harmonic is $f_3 = 3f_1$ etc.

**EXAMPLE 11.22**

Consider a string in a guitar whose length is 80 cm and a mass of 0.32 g with tension 80 N is plucked. Compute the first four lowest frequencies produced when it is plucked.

***Solution***

The velocity of the wave

$$v = \sqrt{\frac{T}{\mu}}$$